

Week 1 - Discovering the Atmosphere 1: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

IN CLASS Question 1: Choose the observations you actually need

A forecaster has access to the following observing systems:

A	surface thermometer/hygrometer	E	weather radar
B	barometer	F	ceilometer
C	anemometer/wind vane	G	radiosonde
D	rain gauge	H	satellite imager

For each task below, choose the *single most useful* system from the list and justify your choice in one sentence. Where another observing system would provide genuinely useful complementary information, name *one additional system* and state what it adds.

1. Warn Manchester Airport about low cloud and poor visibility.
2. Determine whether a line of heavy showers is approaching Manchester during the next hour.
3. Measure the vertical temperature and humidity structure before a possible convective outbreak.
4. Establish whether surface pressure is falling rapidly as a depression approaches.

YOU SHOULD TRY Question 2: Diagnose poor exposure at an automatic weather station

An automatic weather station has been installed with the following sensor placements.

Variable	Sensor placement
Air temperature	unshielded sensor fixed to a sunlit brick wall
Wind	anemometer 2 m above ground immediately behind a row of trees
Rainfall	tipping-bucket gauge beneath an overhanging branch
Pressure	calibrated electronic pressure sensor inside a ventilated instrument box

For each of the first three measurements, state the likely direction or nature of the bias and give one practical improvement. Why is the pressure measurement less affected by the external exposure problems listed here?

YOU SHOULD TRY Question 3: Why did fast communication transform forecasting?

A mid-latitude weather system is travelling eastwards at 15 m s^{-1} .

1. How far does it travel in 6 hours?
2. A set of observations takes 8 hours to reach a forecasting office. Would those observations still describe the same weather system at the location where they were made? Quantify your answer.
3. Use the calculation to explain why telegraphy was a major development for synoptic forecasting.

YOU SHOULD TRY Question 4: In situ or remote sensing? Active or passive?

Classify each observing system as *in situ* or *remote sensing*. For the remote-sensing systems, also state whether the instrument is mainly active or passive.

1. A radiosonde measuring temperature and humidity as it ascends.
2. A weather radar transmitting microwave pulses and receiving echoes from precipitation.
3. A ceilometer transmitting laser pulses to determine cloud-base height.
4. A satellite measuring naturally emitted infrared radiation from the Earth-atmosphere system.

EXTRA CHALLENGE Question 5: Choose the satellite orbit for the job

For each task, choose either a *geostationary* or *polar-orbiting* satellite and give the single most important reason for your choice.

1. Follow the development of a rapidly intensifying Atlantic storm every 10–15 minutes.
2. Obtain detailed global temperature/humidity soundings, including high latitudes.
3. Monitor the movement of a convective cloud system across western Europe during one afternoon.

Then state one important limitation of geostationary observations at high latitude.

YOU SHOULD TRY Question 6: Accuracy, precision and calibration

Two temperature sensors are tested in a laboratory bath held at exactly 20.0°C .

Sensor A reads 19.1, 19.2, 19.1, 19.2 and 19.1°C .

Sensor B reads 19.7, 20.4, 20.1, 19.8 and 20.0°C .

1. Calculate the mean reading from each sensor.
2. Which sensor is more precise?
3. Which sensor is more accurate overall?
4. What specific problem would calibration attempt to correct for Sensor A?

IN CLASS Question 7: Can a sparse observing schedule miss a short-lived event?

A sharp wind shift associated with a shower line lasts for 20 minutes. A traditional synoptic station reports only at 00, 03, 06, 09 UTC, and so on, whereas a nearby automatic station stores one-minute values. Assume the 20-minute event is equally likely to begin at any time during a 3-hour interval between synoptic reports.

1. What is the probability that one of the 3-hourly synoptic observation times falls within the 20-minute event?
2. Hence, what is the probability that the event occurs entirely between two successive synoptic reports and is missed by that record?
3. The automatic station reports a wind shift between its 10:17 and 10:18 observations. To approximately what precision can the passage time be identified from this record?

4. Explain why temporal resolution should be chosen with reference to the timescale of the phenomenon being studied.